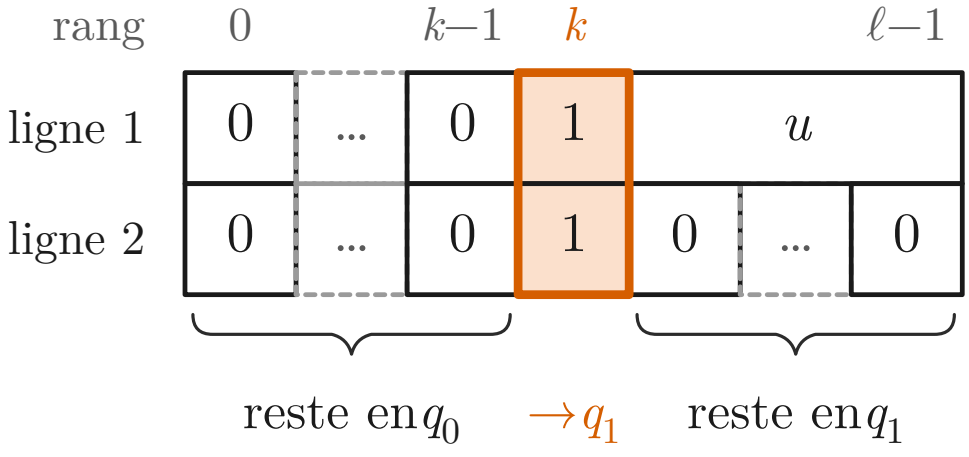

$$v_1(m) = 2^k(1 + 2v(u))$$

$$v_2(m) = 2^k$$